

Understanding Fairness in AI Business Applications: A Case Study on AI Job Interviewing Platforms

LOU THERESE BRANDNER, International Center for Ethics in the Sciences and Humanities, University of Tübingen, Germany

LISA KOERITZ, International Center for Ethics in the Sciences and Humanities, University of Tübingen, Germany

AI-based job interviewing platforms have proliferated in corporate recruitment, marketed not only as efficient alternatives to traditional hiring but as a fairer option: less prone to biases attributed to human decision-makers. Yet these systems have simultaneously drawn criticism for inaccurate and discriminatory outcomes. This mixed-method study examines how companies offering AI-based interviewing tools communicate their commitment to fairness in public-facing materials, investigating how fairness is framed and its relationship to established AI ethics operationalisations. Qualitative analysis identifies 3 dominant communicative strategies, each marked by conceptual shortcomings about the genuine equity of these systems. Quantitative analysis reveals that fairness language is densely and strategically concentrated, consistently positive in sentiment, and anchored to a narrow set of recurring discursive patterns. Together, these findings indicate that the AI hiring industry has developed a relatively coherent fairness vocabulary that serves primarily to legitimate commercial products rather than to substantively address structural fairness challenges. The conceptual integrity of AI fairness must be safeguarded, so that the term retains its critical, normative force rather than being reduced to a rhetorical device.

Keywords: Fairness, AI ethics, corporate communication, automated hiring, mixed methods

Reference Format:

Lou Therese Brandner and Lisa Koeritz. 2026. Understanding Fairness in AI Business Applications: A Case Study on AI Job Interviewing Platforms. In *Proceedings of Fifth European Conference on Algorithmic Fairness (ECAF'26)*. Proceedings of Machine Learning Research, 7 pages.

Hiring and recruitment are time-consuming and costly processes, particularly in large corporations with a high volume of job applicants. It is, thus, unsurprising that the human resources (HR) industry has become a key site of AI-driven automation, with technologies such as AI-supported interviewing platforms increasingly embedded in recruitment and selection processes. Beyond being presented as a cheaper and more efficient option, a key claim made by providers of such systems is that they are *fairer* than traditional methods: they are supposedly less susceptible to bias than human HR professionals, whose decisions can be influenced – consciously or unconsciously – by stereotypes and prejudices [3]. However, AI job interviewing technology has also been publicly criticised for its lack of accuracy and for potentially producing biased results. [21, 22].

Authors' Contact Information: Lou Therese Brandner, International Center for Ethics in the Sciences and Humanities, University of Tübingen, Tübingen, Germany, lou.brandner@uni-tuebingen.de; Lisa Koeritz, International Center for Ethics in the Sciences and Humanities, University of Tübingen, Tübingen, Germany, lisa.koeritz@uni-tuebingen.de.

This paper is published under the Creative Commons Attribution-NonCommercial-NoDerivs 4.0 International (CC-BY-NC-ND 4.0) license. Authors reserve their rights to disseminate the work on their personal and corporate Web sites with the appropriate attribution.

ECAF'26, September 02–September 04, 2026, Ghent, BE

© 2026 Copyright held by the owner/author(s).

Based on this discrepancy, this mixed-method contribution analyses how companies offering AI-based job interviewing systems communicate their commitment to AI fairness to the public. The article incorporates an analytical (computational) social science perspective with normative AI ethics considerations. The aim is to assess how the concept of fairness is framed, both to attract paying customers and to mitigate concerns by applicants and the wider public. The questions that we aim to answer are: How and to what extent is the term *fairness* used as a means within public communication? How does its framing relate to operationalisations in the available literature on AI ethics?

1 Background

AI-supported job interviews have received growing academic attention over the last decade [2, 7–9, 12–14, 20]. AI-supported job interviewing platforms usually conduct one-sided structured behavioural interviews with job applicants without the presence of human recruiters. The interviews typically are recorded in front of a computer camera but can also consist of only audio material. The recorded material is analysed by an AI system, relying largely on language recognition in the shape of speech-to-text transcription and Natural Language Processing (NLP), generating assessments and rankings, on which human recruiters can base further decisions.

All AI systems depend on human input and can thus reproduce existing societal biases and automate them [1, 5, 19]. AI technologies must thus be treated as socio-technical systems embedded in and, in turn, influencing social dynamics. If AI systems in the context of HR procedures make biased recommendations or decisions, population groups such as ethnic minorities or people with disabilities may be more frequently rejected as job applicants, thus perpetuating or even exacerbating existing discrimination in the labour market [7, 11, 18]. Discriminatory decisions can often be traced back to biased training data [15, 19], for instance due to biased labelling and annotation [6]. The term “AI fairness” describes statistical methods designed to mitigate or eliminate social discrimination due to biases in AI models. However, identifying suitable methods is highly context-specific and depends not only on technical issues but also on differing conceptions of fairness and societal value systems, as well as historical and structural power dynamics [10]. Fairness concerns linked to the development and use of automated interviewing have been discussed from ethical and legal perspectives [4, 8, 17, 20].

Corporate websites represent a discursive site through which companies shape public understanding of their products and make implicit or explicit claims about the quality and equity of their systems [16]. Analysing how fairness is framed in this context therefore allows to identify potential gaps between companies’ rhetorical commitments and the structural risks their technologies pose, particularly since such framings may obscure the role of biased AI systems in producing discriminatory outcomes. Understanding these framing strategies can help holding AI hiring companies accountable and informing the broader public discourse on fairness and discrimination in algorithmically mediated hiring.

2 Methodology

This contribution is based on a mixed-method study, combining qualitative content analysis with a quantitative corpus analysis. Our sample was derived from an initial mapping of 50 AI recruiting and hiring companies carried out in 2024 and 2025, from which the focus was narrowed down to 20 companies specifically offering AI-based job interviewing technology. Companies providing other forms of recruitment technology or whose analytical

processes were not explicitly AI-driven were excluded. Since one of the 20 companies (*CandyDate*) was later excluded from the analysis because its website did not provide sufficient text content for analysis, the final sample consisted of 19 companies.¹ Data collection for both the qualitative and the quantitative analysis was restricted to publicly accessible website material, such as website sections, blog posts, and white papers, ensuring that only content directed at general audiences – e.g., prospective corporate clients, job applicants, the wider public – was included.

To address the research questions of how AI job interviewing companies conceptualise algorithmic fairness and how fairness is framed for different audiences, we conducted a qualitative content analysis of company website material. The qualitative analysis employed open coding to identify how fairness is constructed, justified, and communicated across the sampled companies. The selected companies' websites were searched for the terms “fairness”, “fair”, “bias” and “debiasing”. Sections containing these terms were read closely, marking relevant paragraphs and phrases, inductively identifying recurring themes, tone, and messaging patterns without imposing preset categories. Similar codes were iteratively grouped into broader clusters, which were refined through constant comparison until 3 distinct, internally consistent communication strategy types emerged. This inductive strategy allowed recurring themes, framings and conceptual patterns to emerge from the data, enabling a nuanced examination of the discursive strategies through which AI interviewing companies present their system as “fair”.

A quantitative corpus analysis was subsequently conducted to identify broader linguistic patterns and provide empirical grounding to the qualitative findings. The corpus comprised 51 documents from companies already captured in the qualitative sample, with a focus on content addressing fairness and bias. Rather than web scraping, all websites were manually cached and stored locally (28.11.2025) due to the frequency with which company websites were updated, which would have undermined comparability across analytical runs.

The construct of fairness was operationalised across three dimensions: fairness metrics, discourse density, and conceptual proximity. Fairness metric mentions were extracted by using a lexicon based on fairness metrics and running an exploratory presence/absence count to measure how often fairness metrics are discussed in the corpus. Next, discourse density was measured as a normalised frequency metric representing the proportion of fairness-related keywords, such as “fairness”, “unfair”, “bias” and “equity”, relative to each firm's total lexical volume. Conceptual proximity and discursive strategies were examined through a two-stage pipeline in which a deductive regex-based extraction layer first isolated unique fairness-candidate sentences based on pre-defined thematic categories which were based on the three main communicative strategies identified during qualitative analysis (see section “Qualitative Findings”) to create a corpus of fairness-related phrases. Candidate passages were identified at the sentence level, and each match was preserved with contextual evidence snippets and document metadata. This stage yielded 694 unique fairness-candidate sentences. A generative large language model (Qwen3-32B) was then applied to these sentences, evaluating each within a three-sentence rolling contextual window to preserve argumentative context before mapping the corpus deductively against the three predefined thematic frameworks or into an “other/emerging” category.

¹ Alex, CodeSignal, Hireguide, HireVue, Hyring, Incruiter, Interviewer.AI, Interviewstream, Jobma, Knockri, myInterview, Neufast, TalentSelect.ai, Talview, Truffle, Vervoe, VONQ, Willo, XOPA AI

3 Results

3.1 Qualitative Findings

The qualitative analysis identified 3 distinct communicative strategies through which AI job interviewing companies frame the concept of fairness on their public-facing websites. Table 1 summarises the identified strategies with typical terminology used and lists which companies fall into each of the strategies.

The first strategy, which can be summarised as “Recruiters are biased and AI is not”, positions AI systems in explicit contrast to human recruiters, constructing algorithms as inherently neutral and objective counterparts to prejudiced human decision-making. By foregrounding well-documented cognitive biases in human judgement, these companies rhetorically justify substituting human recruiters with automated systems, implying that algorithmic assessment is by definition more objective. This framing, however, is undermined by the extensive body of research demonstrating that AI systems can reproduce and amplify societal biases and that the appearance of objectivity may obscure discriminatory outcomes. With 10 companies, this is the most prevalent communication strategy within the sample.

We can summarise the second strategy as “AI is biased (but not ours)”; here, we can observe a shift from *unbiased* AI to *debiased* AI. This rhetoric acknowledges the broader critique that AI can be biased, but differentiates the company’s own product by emphasising rigorous technical measures allegedly undertaken to mitigate such risks, including debiasing procedures, fairness audits and ongoing model validation. While this approach signals a degree of critical awareness regarding AI fairness concerns, it frequently overstates the effectiveness of current debiasing methodologies, which remain technically contested and have been shown to reduce bias along some dimensions while introducing or preserving it along others. Furthermore, such claims are rarely substantiated with independently verifiable evidence, making them difficult to assess, particularly from the perspective of job applicants. 4 of the analysed companies fall into this communication strategy.

The third strategy, summarised as “Human–AI synergy”, frames fairness as an outcome best achieved through the combination of human and artificial intelligence, emphasising the status of AI as a decision-support tool that enhances human judgement rather than replacing it. There is an emphasis on transparency and avoiding “black boxes”, so that AI systems can be effective “copilots” for human actors. This communication strategy often incorporates parts of both of the other strategies: since both AI and humans can be biased and have different strengths and weaknesses, they work best complementing each other. While this framing appears more epistemically cautious, it carries its own shortcomings: it may place an unrealistic burden on individual human actors to identify and correct instances of algorithmic discrimination. This might lead to much of the responsibility being placed on individuals, such as hiring managers, who might lack both technical expertise and institutional power to oversee and address concerns. 5 of the analysed companies predominantly use this communication strategy.

3.2 Quantitative Findings

The quantitative corpus analysis largely corroborated and extended the qualitative findings:

The communicative strategy of “Recruiters are biased and AI is not” was found to be the primary strategy identified within the total text corpus, with “AI is biased (but not ours)” and “Human-AI synergy” following.

While all companies in the sample discussed fairness to some extent with keyword mentions ranging from 1 to 32 and an average of 11 per company, the analysis revealed a disparity between the breadth and the depth of this

Table 1. Identified communication strategies

Strategy	Terminology used (examples)	Companies
“Recruiters are biased and AI is not”	“Eliminating”/“removing”/“minimising” “human bias”, “human error”, “unconscious bias”, “subjective interpretations”, “objective data”, “data-driven criteria”, “bias-free data”, “more inclusive hiring”, “bias-free candidate assessment”, “fairness built in”, “based on merit”, “humans are inherently biased”, “bias is easier to fix in AI than in people”	Alex Hyring Incruter Knockri myInterview (“Taira”) Talview (“Ivy”) Truffle Vervoe VONQ XOPA AI
“AI is biased (but not ours)”	“Bias mitigation, “debiased AI“, “guidelines“, “principles“, “testing“, “validation“, “monitor, detect, and mitigate bias”; “embrace ethical AI and data ethics”, “ethical development”, “backed by science”, “academic research”	HireVue Jobma Neufast TalentSelect.ai
“Human–AI synergy”	“Augmentation“ / “support“ instead of “replacement“, “AI as a copilot“, “complement your skills”, “hiring is human”, “AI-powered, human-centric”, “humans make the decisions”, “human-in-the-loop”, “human oversight”, “the human touch”, “transparency”, “avoid black boxes”, “explainable AI”	CodeSignal Hireguide Interviewer.ai Interviewstream Willo

engagement. Only 4 out of 19 companies (21%) explicitly referenced concrete fairness metrics, while the remaining 79% recorded zero mentions of specific fairness concepts, suggesting that the majority of companies might engage with fairness more at a rhetorical level than at a technical one. Among those that did discuss measurable fairness standards, engagement varied considerably: The company *HireVue* stood out as the most comprehensive, referencing “algorithmic fairness” across 13 distinct contexts and additionally mentioning “disparate impact”, amounting to 14 metric-specific instances in total, while other companies limited such references to one or two mentions. These findings indicate that although fairness functions as a prominent discursive concept across the industry, companies overwhelmingly prefer broad, conceptual invocations of fairness over engagement with concrete, measurable standards or accountability frameworks. This of course does not directly imply, strictly speaking, that companies do not internally engage with fairness at all. Speaking to the scope of this approach, though, it does show how fairness as a concept is used to promote, support, and ground a complex issue in the public facing context without much of its technical depth. This pattern, however, is consistent with prior literature suggesting that fairness remains a largely surface-level concern in corporate AI communication and raises questions about the gap between public-facing fairness claims and the depth of technical and ethical scrutiny actually applied to these systems.

4 Limitations

As especially the quantitative analysis is focused on identifying the rhetorics of fairness in public facing communication in a limited corpus and the specific context of AI hiring companies, the analysis is still facing several limitations being an exploratory analysis. Further technical validation would improve the approach.

It is also important to emphasise that the qualitative classification into 3 distinct communication types is not absolute or perfect; while the assessed companies could be classified into these three types with relative ease by analysing publicly available material, there can of course be instances of companies mixing communication strategies, utilising rhetorical devices that fall more into another strategy, etc. Our classification represents the *dominant* communication strategy uses within the analysed material.

5 Conclusion

Our findings illustrate that while fairness has become a prominent signifier in the marketing language of AI hiring companies, its deployment is predominantly rhetorical rather than substantive. The three dominant communicative strategies identified each carry conceptual shortcomings that risk misleading clients and applicants about the actual equity of these systems. The quantitative analysis further corroborates this, demonstrating that fairness language is densely and strategically concentrated, framed in consistently positive sentiment and anchored to a narrow set of recurring discursive patterns.

Together, these findings suggest that the AI hiring industry has developed a relatively coherent vocabulary of fairness that functions more to legitimate commercial products than to meaningfully engage with the structural challenges of algorithmic discrimination. This signifies a need to safeguard the conceptual integrity of AI fairness, particularly in high-risk contexts such as hiring decisions. The term should retain a critical, normative meaning as a standard against which systems are evaluated, instead of being hollowed out into a malleable rhetorical device.

Acknowledgments

This article is partly based on research conducted in the projects “FAIR” (Baden-Württemberg State Ministry of Economic Affairs, Labour and Tourism, Germany, funding number BW1_4056) and “ANKER” (German Federal Ministry of Research, Technology and Space (BMFTR), funding number 16SV9223). The authors are responsible for the content of this publication.

References

- [1] Solon Barocas, Moritz Hardt, and Arvind Narayanan. 2023. *Fairness and machine learning: Limitations and opportunities*. MIT press.
- [2] Lou Therese Brandner, Philipp Mahlow, Anna Wilken, Annika Wölke, Hazar Harmouch, and Simon David Hirsbrunner. 2023. How Data Quality Determines AI Fairness: The Case of Automated Interviewing.. In *EWAF*.
- [3] Tomás Chamorro-Premuzic and Reece Akhtar. 2019. Should companies use AI to assess job candidates. *Harvard Business Review* 17 (2019).
- [4] Zhisheng Chen. 2023. Ethics and discrimination in artificial intelligence-enabled recruitment practices. *Humanities and social sciences communications* 10, 1 (2023), 567.
- [5] Virginia Eubanks. 2018. *Automating inequality: How high-tech tools profile, police, and punish the poor*. Macmillan+ ORM.
- [6] Luke Haliburton, Jan Leusmann, Robin Welsch, Sinkar Ghebremedhin, Petros Isaakidis, Albrecht Schmidt, and Sven Mayer. 2025. Uncovering labeler bias in machine learning annotation tasks. *AI and Ethics* 5, 3 (2025), 2515–2528.

- [7] Kimberly Houser. 2019. Can AI solve the diversity problem in the tech industry? Mitigating noise and bias in employment decision-making. *Kimberly A. Houser, Can AI Solve the Diversity Problem in the Tech Industry* (2019).
- [8] Anna Lena Hunkenschroer and Christoph Luetge. 2022. Ethics of AI-enabled recruiting and selection: A review and research agenda. *Journal of business ethics* 178, 4 (2022), 977–1007.
- [9] Alexis Shore Ingber and Nazanin Andalibi. 2025. Emotion AI in Job Interviews: Injustice, Emotional Labor, Identity, and Privacy. In *Proceedings of the 2025 ACM Conference on Fairness, Accountability, and Transparency*. 1–17.
- [10] Jean-Marie John-Mathews, Dominique Cardon, and Christine Balagué. 2022. From reality to world. A critical perspective on AI fairness. *Journal of Business Ethics* 178, 4 (2022), 945–959.
- [11] Stefanie K Johnson, David R Hekman, and Elsa T Chan. 2016. If there’s only one woman in your candidate pool, there’s statistically no chance she’ll be hired. *Harvard Business Review* 26, 04 (2016), 1–7.
- [12] Brittany Kammerer. 2022. Hired by a Robot: The Legal Implications of Artificial Intelligence Video Interviews and Advocating for Greater Protection of Job Applicants. *Iowa Law Review* 107, 2 (2022), 817.
- [13] Jin-Young Kim and WanGyu Heo. 2022. Artificial intelligence video interviewing for employment: perspectives from applicants, companies, developer and academicians. *Information Technology & People* 35, 3 (2022), 861–878.
- [14] Alina Köchling, Shirin Riazzy, Marius Wehner, and Katharina Simbeck. 2020. Highly accurate, but still discriminatory: A fairness evaluation of algorithmic video analysis. In *Academy of Management Proceedings*, Vol. 2020. Academy of Management Briarcliff Manor, NY 10510, 13339.
- [15] David Leslie. 2020. Understanding bias in facial recognition technologies. *arXiv preprint arXiv:2010.07023* (2020).
- [16] Irene Pollach. 2003. Communicating corporate ethics on the world wide web: a discourse analysis of selected company web sites. *Business & Society* 42, 2 (2003), 277–287.
- [17] Manish Raghavan, Solon Barocas, Jon Kleinberg, and Karen Levy. 2020. Mitigating bias in algorithmic hiring: Evaluating claims and practices. In *Proceedings of the 2020 conference on fairness, accountability, and transparency*. 469–481.
- [18] Candice Schumann, Jeffrey Foster, Nicholas Mattei, and John Dickerson. 2020. We need fairness and explainability in algorithmic hiring. In *International conference on autonomous agents and multi-agent systems (AAMAS)*.
- [19] Andrew D Selbst. 2017. Disparate impact in big data policing. *Georgia law review* 52, 1 (2017), 6.
- [20] Nancy T Tippins, Frederick L Oswald, and S Morton McPhail. 2021. Scientific, legal, and ethical concerns about AI-based personnel selection tools: A call to action. *Personnel Assessment and Decisions* 7, 2 (2021), 1.
- [21] Sheridan Wall and Hilke Shellmann. 2021. We tested AI interview tools. Here’s what we found. *MIT Technology Review* (2021).
- [22] Rachel Withers. 2020. Should Robots Be Conducting Job Interviews? *Slate* (2020).